Experiment 6: Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

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Aim

To automate business processes in the "Gym Membership Enrollment & Front Desk Management" application using Deluge scripting and workflows in Zoho Creator, and to test and debug the application.

Application Details

- **Application Name:** Gym Membership Enrollment & Front Desk Management
- **Form:** Membership_Enrollment_Form
- **Report:** Membership_Enrollment_Report
- **Users:** Prospective members, Front Desk and Trainers

Workflows Implemented

S.No	Workflow	Type	Trigger	Purpose
1	Validations on form submission	Form workflow	Create, On Validate	Checks age range, past start date and waiver acceptance before saving
2	Email information	Form workflow	Create, On Success	Sends a confirmation email to the member
3	Front desk notification	Form workflow	Create, On Success	Alerts the front desk about each new enrollment
4	Joining day reminder	Schedule	Daily, 8:00 AM	Reminds members whose start date is today

Procedure

1. Opened the application and went to **Workflow, Form Workflows, New Workflow**.
2. Selected Membership_Enrollment_Form and chose the event (On Validate or On Success).
3. Selected the action condition **Always** and the action type **Deluge script**.
4. Wrote and saved the script for each workflow.
5. Created the reminder under **Workflow, Schedules** with a daily frequency.
6. Tested every workflow using the live form and corrected the errors found.

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Deluge Scripts

1. Validation Workflow (On Validate)

```
if (input.Age < 14 || input.Age > 75)
{
    alert "Age must be between 14 and 75 years.";
    cancel submit;
}

if (input.Preferred_Start_Date < zoho.currentdate)
{
    alert "Preferred Start Date cannot be in the past.";
    cancel submit;
}

if (input.Health_declaration_waiver.isEmpty())
{
    alert "You must accept the Health Declaration & Waiver to enroll.";
    cancel submit;
}
```

2. Confirmation Email (On Success)

```
sendmail
[
    from: zoho.adminuserid
    to: input.Email_address
    subject: "Welcome to the Gym, " + input.Member_Name.first_name + "!"
    message: "Hi " + input.Member_Name.first_name + " " + input.Member_Name.last_name
+ ",<br><br>"
           + "Thank you for enrolling. Here are your details:<br>"
           + "<b>Plan:</b> " + input.Membership_Plan + "<br>"
           + "<b>Activities:</b> " + input.Preferred_activities.toString() + "<br>"
           + "<b>Start Date:</b> " +
input.Preferred_Start_Date.toString("dd-MMM-yyyy") + "<br>"
           + "<b>Workout Slot:</b> " + input.Preferred_workout_slot.toString("hh:mm
a") + "<br><br>"
           + "Please visit the front desk with a valid ID on your start
date.<br><br>Regards,<br>Front Desk Team"
]
```

3. Front Desk Notification (On Success)

```
sendmail
[
    from: zoho.adminuserid
    to: "your-email@example.com"
    subject: "New enrollment: " + input.Member_Name.first_name + " " +
input.Member_Name.last_name
    message: "A new member has enrolled.<br><br>"
           + "<b>Plan:</b> " + input.Membership_Plan + "<br>"
           + "<b>Type:</b> " + input.Membership_Type + "<br>"
           + "<b>Activities:</b> " + input.Preferred_activities.toString() + "<br>"
           + "<b>Start Date:</b> " +
```

```
input.Preferred_Start_Date.toString("dd-MMM-yyyy") + "<br>"
          + "<b>Workout Slot:</b> " + input.Preferred_workout_slot.toString("hh:mm
a") + "<br>"
          + "<b>Contact:</b> " + input.Contact_Number
];
```

4. Joining Day Reminder (Schedule, Daily)

```
today = zoho.currentdate.toDate();
all_members = Membership_Enrollment_Form[ID != 0];

for each member in all_members
{
    if (member.Preferred_Start_Date.toDate() == today)
    {
        sendmail
        [
            from: zoho.adminuserid
            to: member.Email_address
            subject: "Your gym membership starts today!"
            message: "Hi " + member.Member_Name.first_name + ", your " +
member.Membership_Plan + " starts today. See you at " +
member.Preferred_workout_slot.toString("hh:mm a") + "!"
        ]
    }
}
```

Testing

Test Case	Input	Expected Result	Actual Result
Invalid age	Age = 10	Alert shown, submission blocked	Passed
Past start date	Start date = yesterday	Alert shown, submission blocked	Passed
Waiver unchecked	Checkbox empty	Alert shown, submission blocked	Passed
Valid submission	All fields valid	Record saved, welcome email and front desk email sent	Passed
Scheduled reminder	Start date = today	Reminder email received	Passed

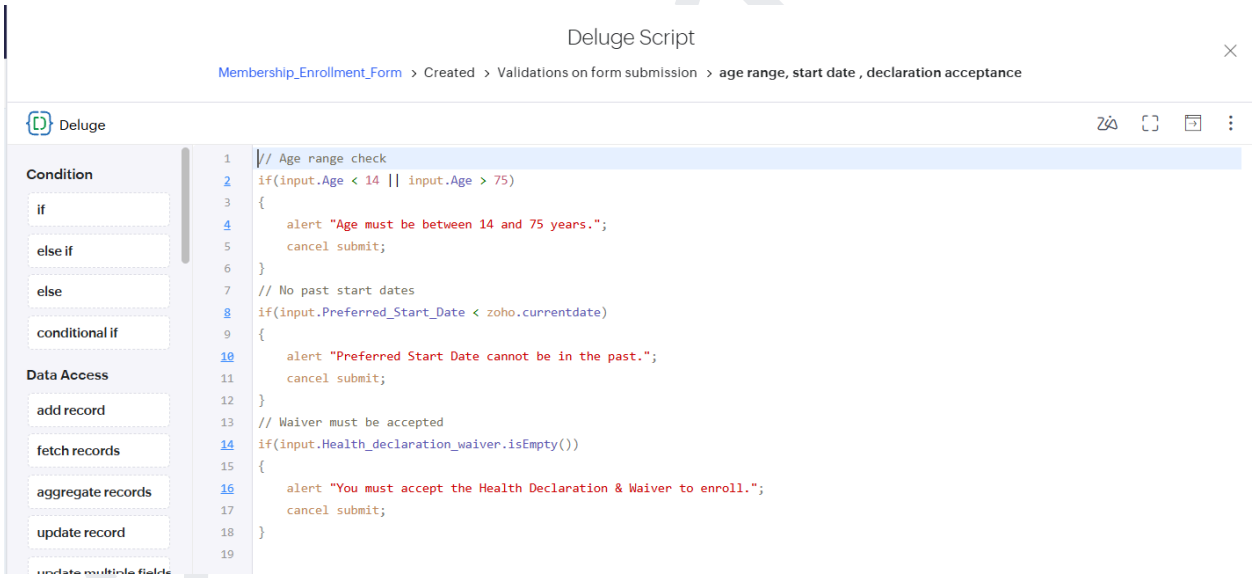
Debugging

S.No	Error Message	Cause	Fix
1	Left expression is of type STRINGLIST and the operator == is not valid	The Checkbox field returns a list, so it cannot be compared with false	Used isEmpty() to check whether the box is ticked

S.No	Error Message	Cause	Fix
2	Variable 'Preferred_Workout_Slot' is not defined	Field link names are case-sensitive; the correct name is Preferred_workout_slot	Used the exact link names from the Fields panel
3	Arithmetic operations not allowed on TIME values	A TIME value was joined to text with +	Converted it using toString("hh:mm a")
4	Improper Statement Error at line 2	Syntax problem in the sendmail block (comment inside the block and missing semicolon)	Removed the comment and ended the statement with ;
5	Reminder firing on every submission	The reminder was placed under On Success instead of a schedule	Moved it to a daily Schedule and removed it from the form workflow

Screenshots

1. Validation workflow script



2. Confirmation email workflow script



3. Front desk notification script



4. Schedule configuration and script

Deluge Script

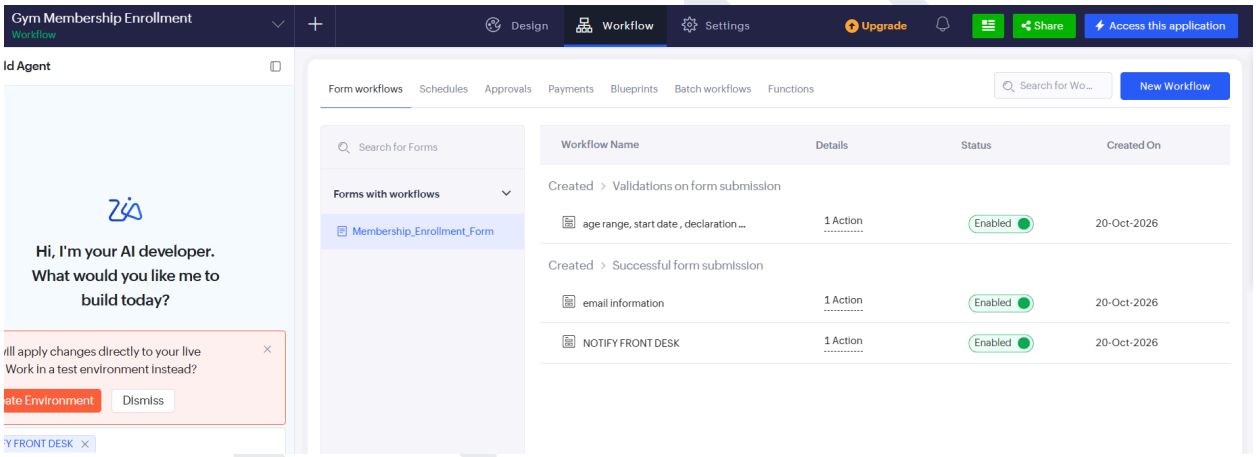
Membership_Enrollment_Form > Based on Preferred Start Date > **EMAIL REMINDER TO JOIN**

Deluge

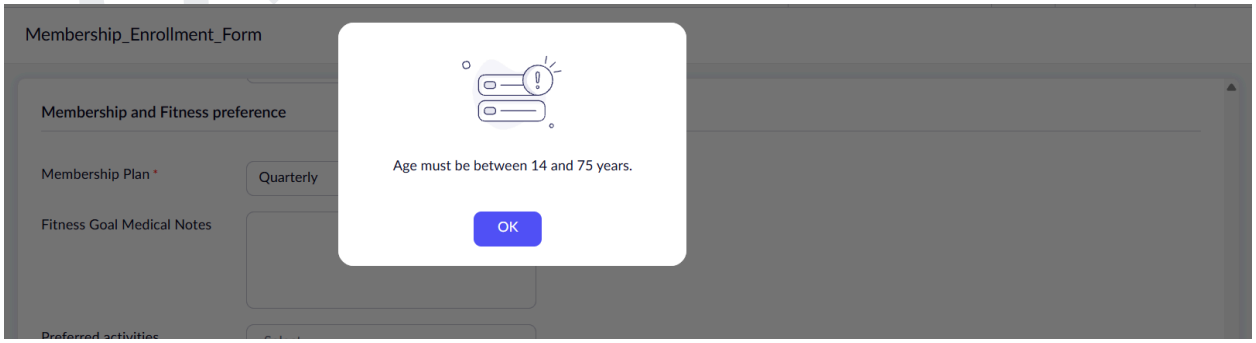
Notifications
send mail
send report
Condition
if
else if
else
conditional if
Data Access
add record

```
today = zoho.currentdate.toDate();
all_members = Membership_Enrollment_Form[ID != 0];
for each  member in all_members
{
    if(member.Preferred_Start_Date.toDate() == today)
    {
        sendmail
        [
            from :zoho.adminuserid
            to :member.Email_address
            subject :"Your gym membership starts today!"
            message :"Hi " + member.Member_Name.first_name + ", your " + member.Membership_Plan + " starts today. See you at " +
member.Preferred_workout_slot.toString("hh:mm a") + "!"
        ]
    }
}
```

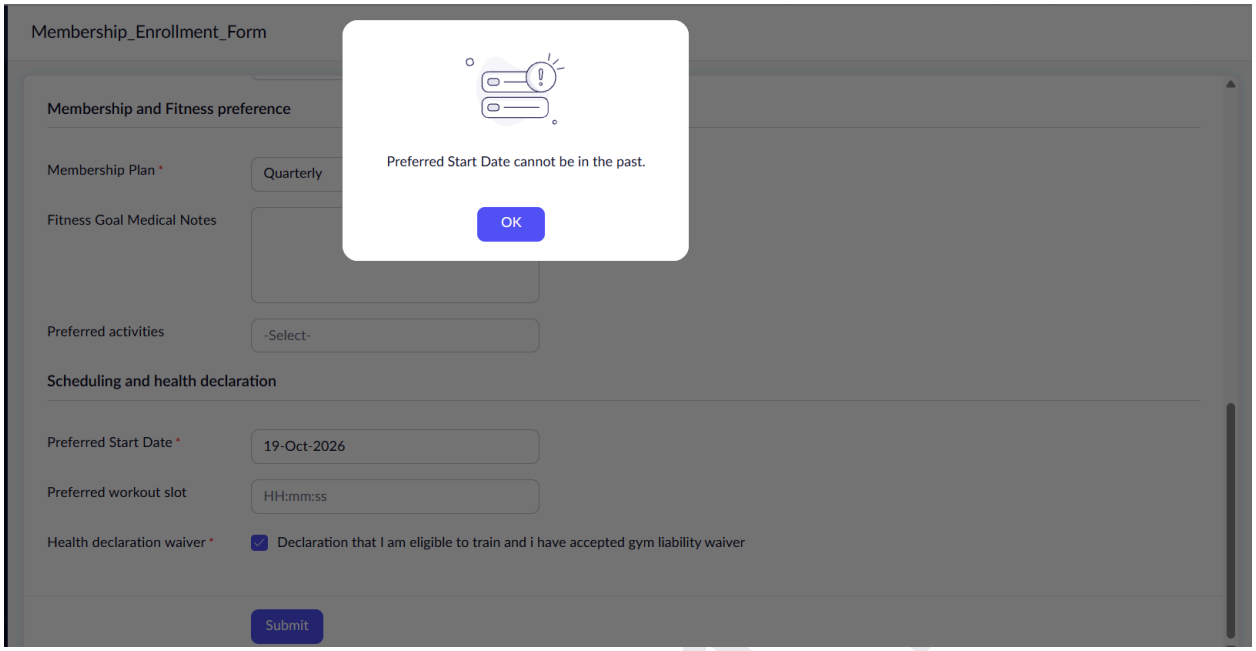
5. Form workflows list showing all workflows enabled



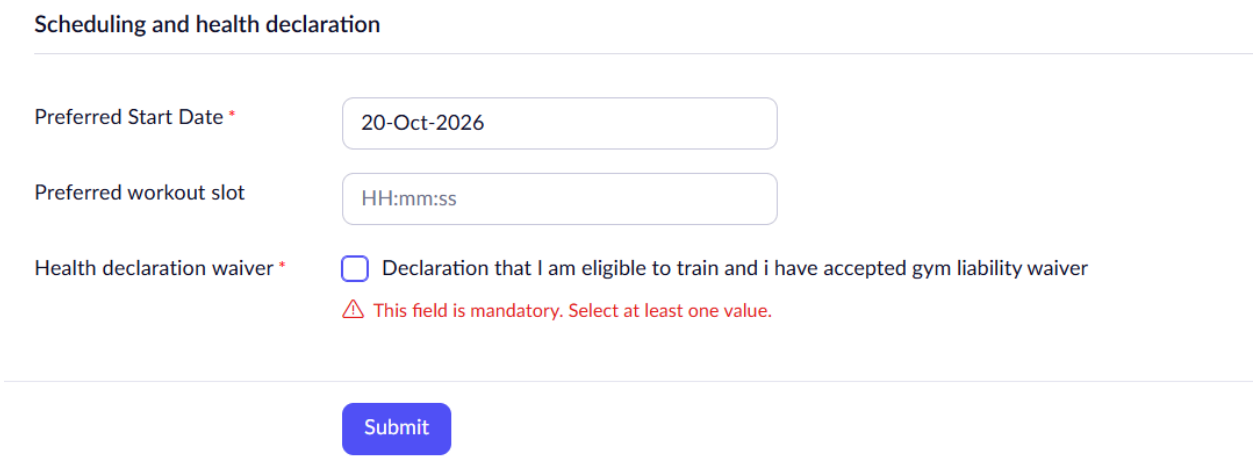
6. Blocked submission: age validation alert



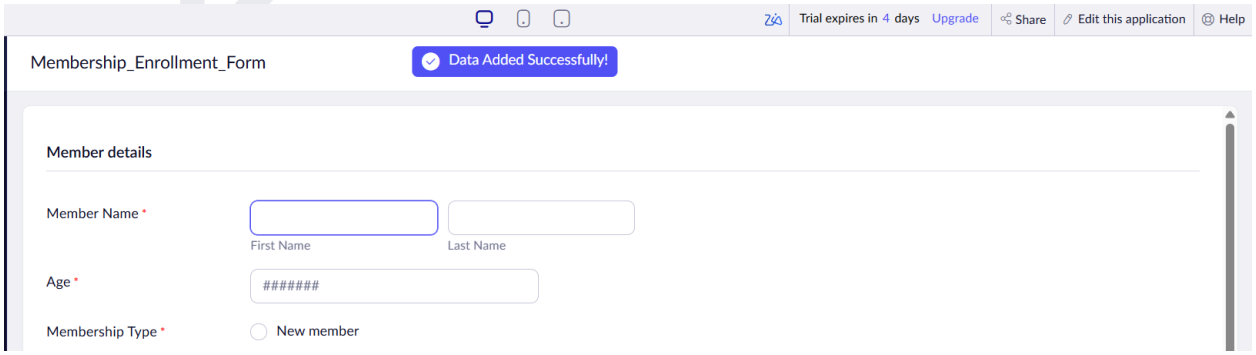
7. Blocked submission: past date alert



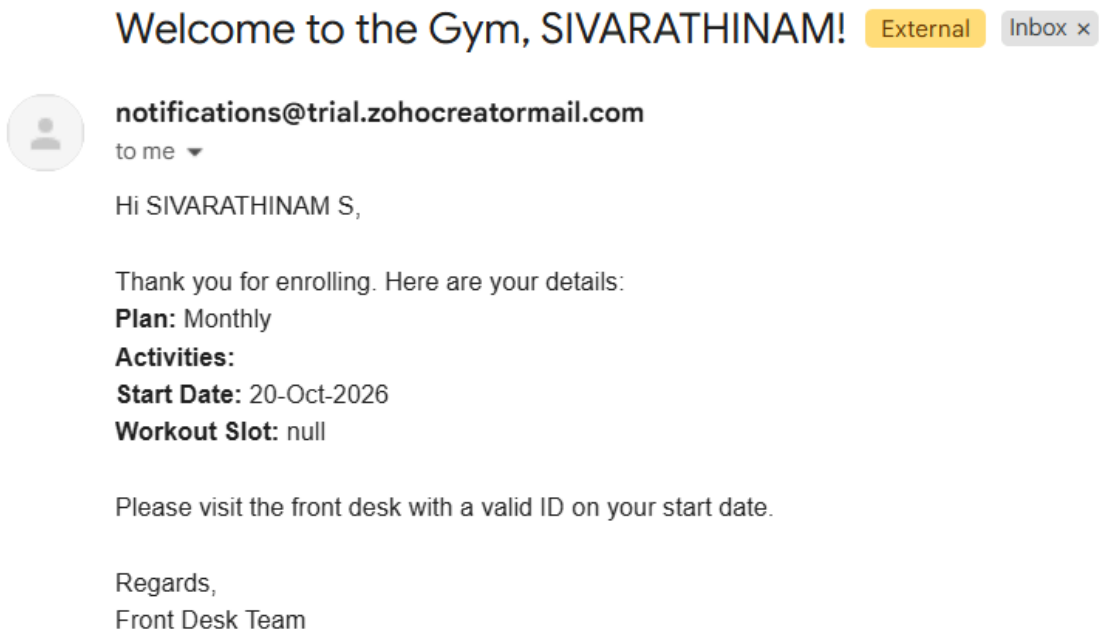
8. Blocked submission: waiver alert



9. Successful submission and report record



10. Confirmation email received by the member



Welcome to the Gym, SIVARATHINAM! External Inbox ×

notifications@trial.zohocreatormail.com
to me

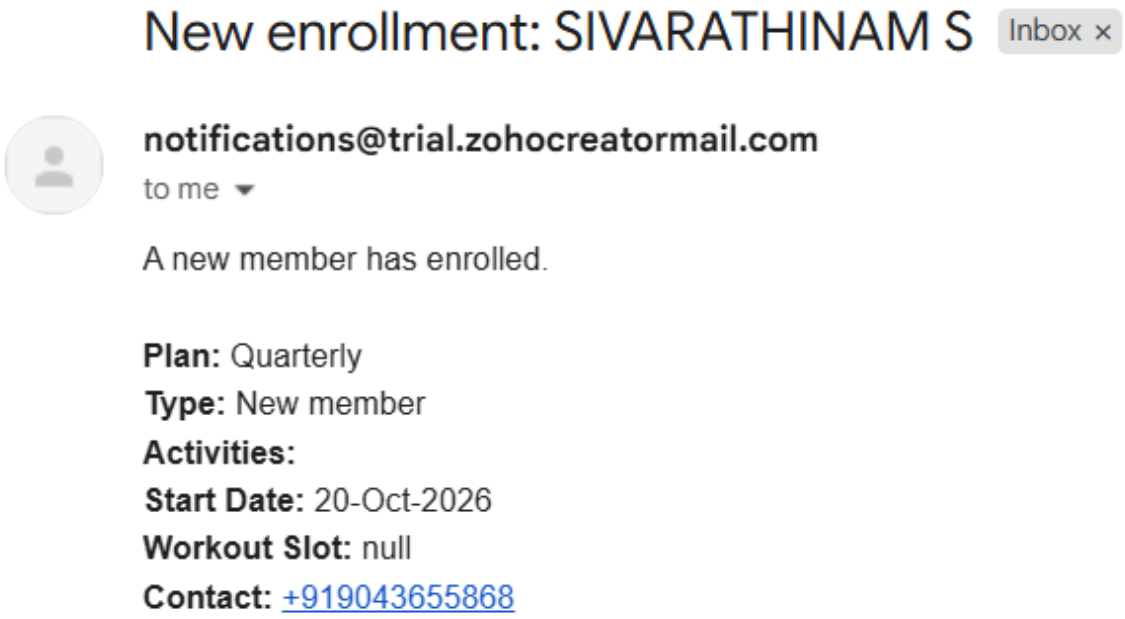
Hi SIVARATHINAM S,

Thank you for enrolling. Here are your details:
Plan: Monthly
Activities:
Start Date: 20-Oct-2026
Workout Slot: null

Please visit the front desk with a valid ID on your start date.

Regards,
Front Desk Team

11. Front desk notification email received



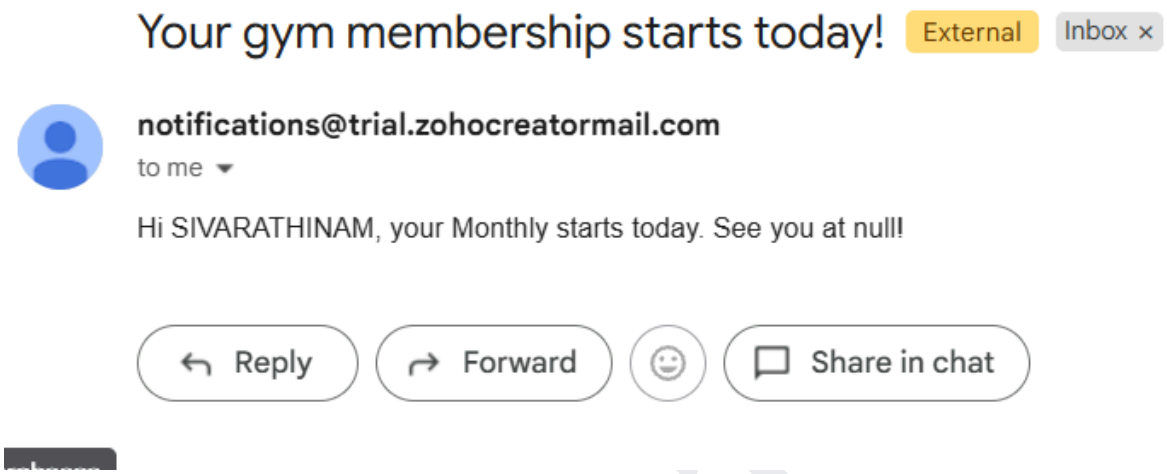
New enrollment: SIVARATHINAM S Inbox ×

notifications@trial.zohocreatormail.com
to me

A new member has enrolled.

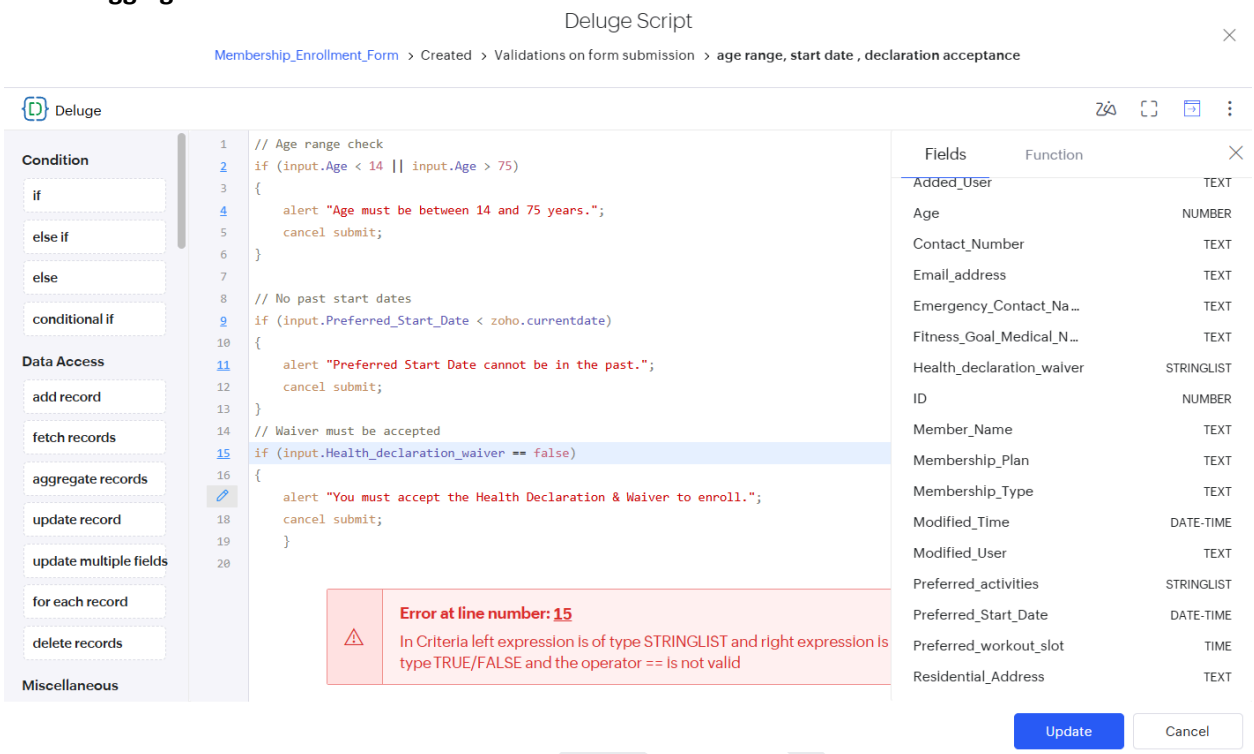
Plan: Quarterly
Type: New member
Activities:
Start Date: 20-Oct-2026
Workout Slot: null
Contact: +919043655868

12. Scheduled reminder email received



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13. Debugging errors and their fixes



Result

The validation, confirmation email, front desk notification and scheduled reminder workflows were created using Deluge scripting in Zoho Creator. The application was tested with valid and invalid inputs, the errors found were debugged, and the business process automation was completed successfully.

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